Hydrology Research

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Hydrology Research Vol 57 No 4, 804 doi: 10.2166/nh.2026.236

Machine learning-based streamflow forecasting in Colombian hydropower basins at daily and monthly scales

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ABSTRACT

This study evaluates three machine learning methods: Random Forest (RF), Long Short-Term Memory networks (LSTM), and Multilayer Perceptron Artificial Neural Networks, to forecast streamflow at daily and monthly scales for the Betania, Quimbo, Hidrosogamoso, and Urrá hydropower basins in Colombia. Model skill was assessed using two input configurations: (1) temperature and rainfall with multiple time delays, and (2) the same predictors including past inflow. Results indicate that no single model, configuration, or temporal resolution performs best across all basins. In some catchments, meteorological variables alone are sufficient, while in others, antecedent conditions represented by previous inflow are essential to improve accuracy. Monthly aggregation can enhance peak-flow representation but may increase overfitting, especially in basins with limited data; thus, daily data often remain more informative. During El Niño -Southern Oscillation (ENSO) events, forecasting skill is satisfactory for normal and low flows, whereas performance deteriorates for high-flow conditions. Overall, model performance tends to be better during El Niño than La Niña conditions. These findings support improved strategic energy planning in hydropower-dependent systems, where accurate inflow forecasts are critical for reliability.

Key words: Artificial Neural Networks, Long Short-Term Memory Networks (LSTM), Magdalena River, Random Forest, Sinú River Sogamoso River

HIGHLIGHTS

- Machine Learning models show basin-dependent inflow predictability.
- NSE values above 80–90% can be achieved for some basins with good generalization.
- Temperature and precipitation are sufficient as inputs in some basins.
- Monthly data risks overfitting; daily inputs remain more robust.

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GRAPHICAL ABSTRACT

Input data	Models	Temporal Scales	Key results
Precipitation 	Ramdon Forest 	Daily 	No single model, input configuration, or temporal scale performs best across all basins
Temperature 	Multilayer Perceptron 	Monthly 	Meteorological variables alone are sufficient in some basins, but including past inflow significantly improves accuracy in others, especially for daily forecasting.
Lagged Inflow (Optional) 	Long-Short Term Memory 		Models show good skill for low and normal flows, particularly during El Niño, but poor performance for high flows, especially during La Niña

INTRODUCTION

Water resources are fundamental to human well-being, as they sustain essential needs such as drinking water and sanitation, agriculture, and the majority of economic activities. Because water has multiple uses, it is imperative to plan its distribution strategically. Forecasting future water availability is crucial for this task, especially in light of climate change, increasing water scarcity, and floods, all of which endanger human and ecological well-being. Electricity generation is among the economic activities most susceptible to hydrological variability. Water is essential for both cooling thermoelectric power plants and for driving hydropower generation. Therefore, accurate streamflow prediction is especially important for the planning, operation, and optimization of hydroelectric facilities. Reliable forecasts enable operators to anticipate electricity production, manage reservoir storage, and mitigate risks associated with floods or droughts. For forecasting water availability, precipitation and temperature are crucial predictors, as rainfall directly generates runoff and temperature has a direct influence on evapotranspiration; both are key components of the hydrological cycle.

For predicting streamflow in hydropower basins, physically based models, despite their theoretical advantages, are limited by their complexity, extensive data requirements, dependence on expert knowledge and computational resources, and challenges related to translations between spatial scales and complex calibration (Özdoğan-Sarıkoç & Dadaser-Celik 2024). As an alternative, data-driven hydrological methods have been increasingly adopted for streamflow forecasting. Recent directions in hydrological forecasting include hybrid physical-ML approaches that integrate physical constraints into machine learning architectures to improve consistency, as well as large-sample and regional learning frameworks that allow models to generalize across multiple catchments (Wilbrand *et al.* 2023; Jahanbani *et al.* 2024). Within these frameworks, Long Short-Term Memory (LSTM) has become an important advancement since it can show long-term hydrological memory (Kratzert *et al.* 2018; Nie *et al.* 2025). However, approaches such as Random Forest (RF) and Multilayer Perceptron (MLP) remain very relevant for operational forecasting due to their computational efficiency and performance at different time scales (Pham *et al.* 2021; Jesus *et al.* 2025).

Despite these global advancements, applying ML methods in tropical regions faces two fundamental challenges. First, tropical basins are highly diverse, exhibiting significant hydroclimatic heterogeneity; moreover, their hydrological responses to climate change are expected to differ substantially, resulting in divergent sensitivities that pose significant challenges for generalized or transferable hydrological and machine-learning models (He *et al.* 2024). Second, ML architectures typically require large, high-quality datasets to avoid overfitting and ensure generalizability (Oneto *et al.* 2025). However, many

tropical basins are characterized by significant data scarcity and temporal gaps in hydrometeorological variables that are costly and demanding to observe, such as evapotranspiration and soil moisture (Seneviratne *et al.* 2010; World Meteorological Organization and Alliance for Hydromet Development 2024). Under these limitations, there is a gap in the literature on how different ML architectures behave when given a minimal set of predictors, such as precipitation and temperature. This is important, as a few but relevant predictors may in fact avoid overfitting and improve out-of-sample model performance (Akbarian *et al.* 2023). Then, evaluating the performance of parsimonious models in these data-limited settings is critical for developing low-cost, accurate forecasting tools.

Furthermore, the El Niño–Southern Oscillation (ENSO) adds an additional layer of complexity. This ocean–atmosphere phenomenon oscillates between two contrasting phases, El Niño and La Niña, which typically induce significant temperature and precipitation anomalies. In tropical regions, ENSO plays a crucial role in interannual climate variability. Colombia serves as an ideal case study since it relies heavily on hydropower and is impacted by ENSO. In this country, these events significantly affect rainfall, river flows, and reservoir storage, leading to extended droughts during El Niño and unusually high flows during La Niña. Consequently, models must simultaneously forecast these opposing extremes.

This study utilizes hydrometeorological data from four Colombian hydropower basins representing diverse hydroclimatic conditions of the country. In these systems, rainfall and temperature act as key climatic drivers of runoff generation and streamflow variability, modulated by large-scale phenomena such as ENSO. Therefore, the objective of this study is to evaluate and compare the performance of RF, MLP, and LSTM models for streamflow forecasting in the context of ENSO across a diversity of hydroclimatic conditions in Colombia.

STUDY AREA AND DATA

Case studies

Four hydropower plants (Central Hidroeléctrica Urrá, Central Sogamoso, Central Hidroelectrica Betania, and Central Hidroeléctrica El Quimbo) in Colombia were selected as case studies (Figure 1) due to their relevance in Colombia's national energy system and contrasting hydrological and operational conditions. These hydropower plants are located in distinct climatic regions, which provide a diverse range of inflow dynamics.

Central Hidroeléctrica Urrá (CHU) is situated at the southern end of the Córdoba department on Colombia's Caribbean Coast. It is the sole hydropower facility in this area, served primarily by the Sinú and Verde rivers, located downstream of the Paramillo National Natural Park. The rainfall pattern features one dry season from November to April and a wet season from May to early November. The basin experiences a humid tropical climate, with an average of 1,100–2,000 mm of rain falling in this region per year, high relative humidity and average temperatures between 26 and 30 °C (Marrugo-Negrete *et al.* 2017). The Montería gauging station, downstream of CHU, reports a mean annual discharge of the Sinú river of approximately 400 m^3/s. During the low-flow period (December–April), monthly average discharges range between 50 and 160 m^3/s. In contrast, during the rainy season, the system exhibits markedly higher flow conditions, with mean monthly discharges fluctuating between 578 and 838 m^3/s (Feria Díaz *et al.* 2017). The construction of CHU commenced in 1993, and the plant began operations in 2000. It is equipped with four Francis turbines and has a total installed capacity of 340 MW. The reservoir's total volume is 1,616 million m^3, with a usable volume of 1,213 million m^3, and an average monthly discharge of 334 m^3/s. It serves multiple purposes, including electricity generation, streamflow regulation, fishing, and transportation.

Central Hidroeléctrica Sogamoso (CHS) is located in the Santander department, in the canyon where the Sogamoso river, its tributary, crosses the La Paz mountain range, which is part of the middle Magdalena basin. This region experiences a bimodal rainfall pattern with a first rainy peak in April/May, reaching an average of 200 mm per month, and a second peak in September/October, when rainfall averages 217 mm (Cerón *et al.* 2022). The driest conditions are observed during the December–February (DJF) period, when monthly precipitation drops below 80 mm. A secondary minimum occurs between June and August, during which mean monthly rainfall reaches approximately 140 mm (Cerón *et al.* 2022). The reservoir is located in a warm, dry climate with average temperatures between 27 and 28 °C. The upstream parts of the basin are more humid due to higher elevations. The monthly mean discharge of the Sogamoso river exhibits a bimodal distribution with two annual peaks: the first occurring in May, reaching 775 m^3/s, and the second during October and November, with values of 765 and 776 m^3/s, respectively. The lowest monthly average discharges are recorded in January (200 m^3/s) and February (193 m^3/s) (Ramírez & Pinilla 2012). Construction of the dam began in 2009, and operations

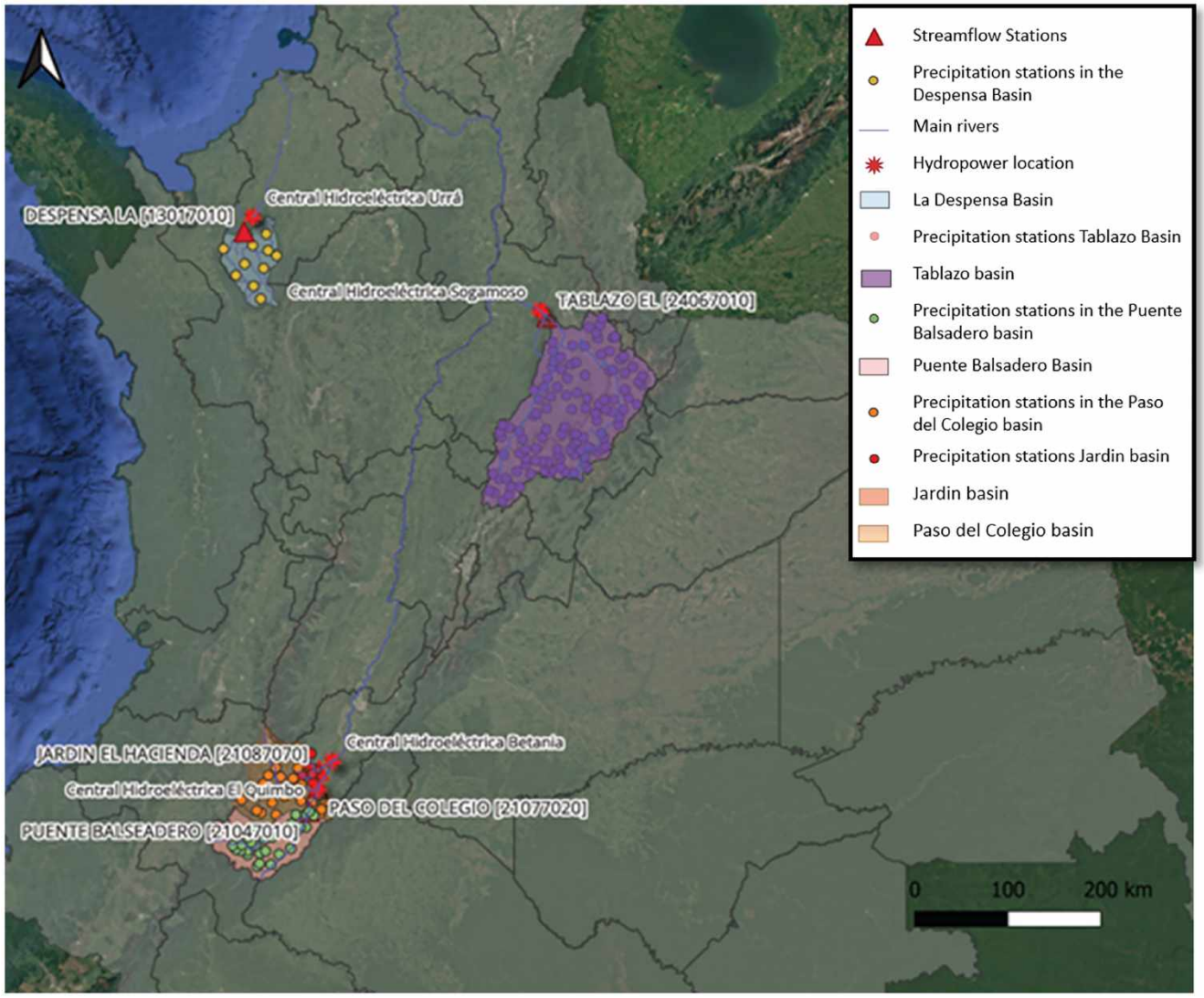


Figure 1 | Location of case study hydropower plants and their respective basins.

started in December 2014. The plant has three Francis turbines and a capacity of 819 MW. The reservoir's total volume is 4,800 million m^3, with a usable volume of 2,975 million m^3 and an average monthly discharge of 455 m^3/s. The reservoir is called Topocoro, and its primary use is electricity generation.

Central Hidroeléctrica Betania (CHB) is located 200 meters downstream of the confluence of the Yaguará and the Magdalena rivers in the El Huila department. Construction started in 1981, and operations began in 1987. The plant has three Francis turbines and an installed capacity of 540 MW. The reservoir has a total volume of 1971 million m^3, and a usable volume of 1,042 million m^3, with an average monthly discharge of 388 m^3/s. The reservoir has multiple purposes, including electricity generation, streamflow regulation, irrigation, and fish farming.

Central Hidroeléctrica El Quimbo (CHQ) is located 12 km upstream of CHB in the El Huila department. Its main tributaries are the Magdalena and Sauza rivers. Construction started in 2010, and operations began in November 2015. The plant has two Francis turbines and an installed capacity of 400 MW. The reservoir has a total volume of 3,205 million m^3 and a usable volume of 2,354 million m^3 with an average monthly discharge of 200 m^3/s. It is a single-purpose, electricity generation reservoir, with some tourism and fishing. Together with CHB, the plant forms a generation chain on the upper Magdalena River.

The upper Magdalena basin has a bimodal precipitation regime. The first, more intense rainy season extends from March to May, with peaks in April and May, and the second rainy season extends from October to November. The mean annual precipitation is approximately 1535 mm, reflecting a monsoonal rainfall regime with pronounced seasonality (Escobar *et al.* 2017). A warm and humid climate characterizes the basin; however, at the CHB reservoir site, it is dry and warm with average temperatures ranging from 23 to 26 °C. The CHQ reservoir site is more humid with an average temperature of 26 °C. The average annual runoff reaches around 900 mm, indicating a marked hydrological response to rainfall pulses. This sector of the basin exhibits high spatial variability in precipitation, primarily driven by Andean orographic effects and seasonal atmospheric circulation patterns (Escobar *et al.* 2017).

Interannual climate oscillations, particularly ENSO, influence the four hydropower plants, but in different ways. In Hidrosogamoso and Betania, ENSO shows a stronger hydrological signal as El Niño leads to marked reductions in rainfall and streamflow, while La Niña increases streamflow and energy generation. In the CHQ basin, inflow responses are weaker, and the reservoir tends to be operated more strategically to anticipate future dry years while Urrá is the least affected by El Niño, with a lesser decrease in electricity generation (Pulgarín-Morales & Krueger 2025). The inflow dynamics of CHB, CHQ, and CHS are additionally shaped by anthropogenic interventions such as upstream flow regulation and land-use changes. Betania, in particular, is regulated by the upstream CHQ reservoir, which significantly alters the basin's natural hydrological response. In contrast, the Sinú basin, where the Urrá hydropower plant is located, is characterized by more direct catchment responses to rainfall, but is affected by extensive agricultural and cattle-ranching activities. This diversity allows for the evaluation of the forecasting models under different hydrological regimes, operational constraints, and catchment response patterns, making the selected set of reservoirs a representative benchmark for streamflow forecasting in tropical hydropower systems.

From an electricity systems perspective, the four reservoirs are vital in Colombia's hydro-dominated energy matrix, especially during El Niño, where accurate inflow prediction is crucial for risk-based decision-making, dispatch optimization, and reservoir operation planning. Testing the forecasting models on both rainfall-influenced systems (CHS, CHU) and multi-reservoir cascade environments (CHB–CHQ generation chain) broadens the assessment to various operational settings. The presence of well-documented historical events like the strong El Niño of 2015–2016 and La Niña of 2021–2022 offers additional opportunities to assess model robustness under extreme hydrological conditions.

Data

The daily mean streamflow (m^3/s), mean air temperature (°C), and precipitation (mm) time series data were acquired from the Institute of Hydrology, Meteorology and Environmental Studies (IDEAM in Spanish). Feature selection was applied as follows to reduce input redundancy and mitigate overfitting, particularly given the limited data availability in some basins. For the streamflow series, the monitoring station closest to each reservoir with the most complete record was selected, and outliers were corrected using data from the nearest gauging station with consistent behavior. This selection ensures that the streamflow record best represents the inflow conditions influencing reservoir operation while minimizing missing data and measurement inconsistencies. Missing streamflow data were not filled to prevent the models from capturing artificial patterns introduced by gap-filling algorithms. In the case of the Urrá basin, inflow records provided directly by the operating company were used, as no official inflow gauging stations are available in this catchment. For precipitation, upper-tail outliers were removed by eliminating all values above the 99.5th percentile to reduce the impact of extreme data point values which tended to be implausible when compared to nearby station records. Missing values were imputed using satellite-based precipitation products, including CHIRPS, CPC, ERA5, ERA5-Land, GPCP, and IMERG. Satellite precipitation was bias-corrected via linear scaling, combined with an Empirical Quantile Mapping correction. This dual approach ensures that the corrected satellite series not only matches the observed mean but also preserves the statistical properties of the local precipitation distribution. The products were evaluated against ground-based rain gauge observations using the Root Mean Square Error (RMSE), the Mean Absolute Error (MAE), and the Mean Absolute Percentage Error (MAPE). The satellite dataset with the lowest error metrics was selected to fill the gaps in the precipitation series.

Temperature was filtered using the same spatial validation criteria as precipitation. Since proximal stations (<2 km) identified both upper and lower tail anomalies that lacked regional coherence, values exceeding the 99.5th percentile or falling below the 0.5th percentile were removed. Data gaps were filled using satellite-derived products such as ERA5, ERA5-Land, GLASS, MOD11A1, MYD11A1, and VNP21A1D. To reduce systematic deviations in temperature, the mean bias between satellite-derived temperature and in-situ observations was applied as an additive correction factor to adjust the satellite

temperature series. Again, the satellite dataset with the lowest error metrics was selected to fill the gaps in the temperature series. Supplementary Data Table S1 provides details on the selected streamflow stations, the number of precipitation and temperature stations in each basin, and the corresponding time-series lengths.

METHODS

The model inputs were time-lagged hydroclimatic predictors, including daily mean air temperature, precipitation, and lagged streamflow values. To reduce input redundancy and retain only the most informative variables, a correlation-based feature selection approach was applied. This approach quantified the strength of association between each candidate predictor and the target discharge, retaining only those with the highest correlation scores (Ateeq-ur-Rauf *et al.* 2018) while preserving interpretability. Before model training, all selected input features were rescaled using min–max normalization to a standardized range of −1 to 1, which improves numerical stability and convergence during training. The output feature was scaled between 0.05 and 0.95 to allow the model to predict greater and smaller inflows than those observed and to avoid saturation effects. To ensure proper generalization and prevent overfitting, the dataset was divided into three subsets: a training set (75%), a validation set (20%), and a test set (5%). This choice was motivated by the need to maximize the amount of data used for model learning and hyperparameter tuning, while still retaining an independent test set to assess generalization on unseen data. The training set is used to learn the model parameters, while the validation set is reserved for tuning hyperparameters and best model selection. The test set provides an additional unbiased estimate of the model's predictive performance by using data that the model has neither seen in training nor in validation. For monthly simulations, several basins (Paso del Colegio, Puente Balsadero, and Despensa) had substantially shorter records. In these cases, the split strategy changed to 67% for training, 23% for validation, and 10% for testing. The models were trained and evaluated on the same dataset to ensure consistent and comparable assessment of predictive skill across basins and algorithms. Four performance metrics, NSE (Nash–Sutcliffe Efficiency), RMSE, MAE, and MAPE, were used, as no single metric is universally reliable in hydrological model evaluation, each one emphasizing different aspects of error. NSE is the most informative for variability and peaks, RMSE further emphasizes large errors, MAE is useful to detect bias, and MAPE is more sensitive to low-flow errors.

To assess model performance under hydrologically critical conditions, two contrasting ENSO phases were selected: an El Niño period (2023–2024) representative of predominantly low-flow conditions, and a La Niña period (2021–2022) characterized by high-flow events. These two events were chosen because consistent records are available across all basins during these periods. This helps to evaluate the robustness of the models under both dry and wet extremes, which are particularly relevant for hydropower operations and risk management. High- and low-flow events were identified using discharge thresholds derived from the full observed streamflow record, corresponding to the 90th and 10th percentiles, respectively. Model performance during each period (La Niña or El Niño) and flow regime (low, normal, and high flows) was evaluated using the Mean Absolute Error (MAE). This provides a stable and interpretable measure of model accuracy when applied to subsets of the flow distribution, avoiding the instability of efficiency metrics under low-variance conditions. This analysis was conducted exclusively at the daily scale, as temporal aggregation to monthly data substantially reduced the number of available observations, limiting the interpretability of the results.

Multilayer Perceptron (MLP)

MLP is an artificial neural network composed of interconnected neurons, mathematically linked, representing a function. The coefficients and intercepts associated with the input variables are referred to as weights and biases, respectively (Equation (1)). The architecture consists of an input layer, one or more hidden layer(s), and an output layer. Information propagates forward through the network from the input layer to the output layer without any feedback connections. Each neuron computes a weighted sum of its inputs and applies a nonlinear activation function in the form of:

$$z_j^l = \sum_{i=1}^{n_{l-1}} w_{ji}^l * a_i^{l-1} + b_j^l \tag{1}$$

where w_{ji}^l is the weight connecting neuron i in layer $l-1$ to neuron j in layer l, b_j^l is the bias term, a_i^{l-1} denotes the activation output from the previous layer and n_{l-1} is the number of neurons in the previous layer. For the first layer n_{l-1} refers to the input data. It follows that z_j^l is the total weighted signal input of neuron j.

The neuron activation output is then obtained by applying a nonlinear activation function $f(\cdot)$ with:

$$a_j^{(l)} = f(z_j^l) \tag{2}$$

The MLP is trained using a backpropagation learning algorithm which minimizes a given loss function through gradient descent:

$$w_{ji}^l \leftarrow w_{ji}^l - \eta \frac{\partial \mathcal{L}}{\partial w_{ji}^l} \tag{3}$$

$$b_j^l \leftarrow b_j^l - \eta \frac{\partial \mathcal{L}}{\partial b_j^l}$$

where η denotes the learning rate, $\mathcal{L}$ is the loss function, $\partial \mathcal{L}/\partial w_{ji}^l$ is the gradient of the loss function with respect to the weight w_{ji}^l, $\partial \mathcal{L}/\partial b_j^l$ is the gradient with respect to the bias b_j^l and $\leftarrow$ refers to an update operator.

Since there is no universally accepted predefined neural network architecture for streamflow prediction, it is necessary to experiment with different combinations of hyperparameters to identify an optimal configuration (Adhikary *et al.* 2017). Therefore, several architectures were tested over the hyperparameter space as detailed in Table S2.

Bayesian optimization was employed to tune the MLP hyperparameters, with an initial random search phase to explore the parameter space before optimization. The best architecture was determined based on validation set performance using the NSE metric (Jeong & Kim 2005). The optimal MLP architecture identified for each basin is presented in Table S5.

Long Short-Term Memory (LSTM)

LSTM was introduced by Hochreiter & Schmidhuber (1997) as a solution to the vanishing and exploding gradient problems that occur during backpropagation in traditional Recurrent Neural Networks. LSTMs have a gated architecture composed of input, forget, and output gates, which regulate the flow of information and allow the network to retain or discard information over long time intervals. This proves useful in modelling dynamic systems with temporal dependencies. The general data flow of an LSTM can be explained through Equations (4)–(9). First, the forget gate f_t and the input gate i_t are computed as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{4}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{5}$$

where x_t is the input vector at time step t (here precipitation and temperature), h_{t-1} is the hidden state vector (the short term memory), W_* and b_* denote the respective weight matrices and the respective bias vectors associated with each gate, and $\sigma(\cdot)$ is a sigmoid activation function, constraining the output values between 0 and 1.

From the same information vectors h_{t-1} and x_t, a candidate cell state $\widetilde{C}_t$ is computed for time step t as:

$$\widetilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{6}$$

which is then used to determine the cell state C_t by multiplication with the input gate i_t:

$$C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t \tag{7}$$

where $\odot$ refers to the Hadamard product, C_{t-1} is the cell state from the previous timestep which is filtered (multiplied) by the forget gate output f_t. The cell state C_t can be thought of as the long-term memory component. The activation function for the candidate cell state $\widetilde{C}_t$ is the hyperbolic tangent $\tanh(\cdot)$ which outputs values in the range $[-1, 1]$.

Finally, the output gate o_t is computed as:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{8}$$

Which then determines the final hidden state h_t for the timestep from the previously calculated cell state C_t (Equation (4)–(7)) as:

$$h_t = o_t \odot \tanh(C_t) \tag{9}$$

At the first time step $t = 1$, the initial hidden state and cell state are initiated as zero vectors and the first input vector x_1 is used to construct the initial internal memory C_1.

A random search optimized the LSTM hyperparameters; the tested hyperparameter space is detailed in Table S3. The optimal architecture identified for each basin is presented in Table S5.

Random Forest (RF)

RF is a nonparametric ensemble learning approach efficient for regression tasks. It works by constructing multiple decision trees using bootstrap aggregation (bagging), where each tree is trained on a resampled subset of the data and a random subset of predictors. At each node split, the algorithm evaluates only this subset and selects the predictor that maximizes the reduction in the variance of the target variable. This randomization reduces the likelihood of dominant predictors being repeatedly selected, which in turn decorrelates the trees and improves generalization performance. Once all trees are trained, the final regression output is obtained by averaging the predictions from all individual trees. This structure reduces variance and improves generalization, making RF highly robust in regression problems with nonlinear relationships and complex interactions between variables (Genuer *et al.* 2017).

Consistent with the previous models, a hyperparameter search was conducted, and the full set of evaluated hyperparameters is provided in the Supplementary Data Table S4. The optimal architecture identified for each basin is presented in Table S5.

Computational hardware

The hyperparameter search and model training were performed using the TensorFlow and Keras deep learning frameworks. All computations were executed on a dedicated machine learning server equipped with three NVIDIA Ada 6,500 GPUs, with 28 GB of dedicated memory each.

RESULTS

This section presents the comparison of the three models. Figures showing time series of observed and predicted daily or monthly streamflow for each basin, using the RF, MLP, and LSTM models under the two input configurations, are provided in the Supplementary Material (Figures S1–S4).

Daily data

Temperature and precipitation as input

This first model configurations use precipitation and temperature as inputs to forecast streamflow into the reservoirs. Table 1 summarizes the performance metrics for the training, validation, and testing phases.

Regarding model architectures, the MLP proved to be the most robust and consistent model. It maintained stable performance metrics across training, validation, and testing phases, particularly in smaller, high-variability basins like Jardín. While the Random Forest (RF) model often achieved the highest accuracy during the training stage, it consistently exhibited signs of overfitting, losing significant predictive power when applied to unseen data. In contrast, the Long Short-Term Memory (LSTM) network showed a particular behavior; while it performed poorly in small or regulated basins, it performed well in large basins, as in Tablazo. This suggests that LSTM's memory gates are most effective in large basins where antecedent conditions and prolonged travel times dominate the discharge response. Additionally, all models except LSTM and MLP in Tablazo struggled to predict peak flows. This is expected as the models in this configuration lack information on antecedent moisture, storage, and other hydrological processes that control runoff generation. The results revealed that the Tablazo basin yielded the best results for the three models compared to the other basins. This can be related to the comparably dense instrumentation of the basin, which enables a good coverage of the entire basin. Moreover, it is a comparably large basin, whose natural hydrological buffering behaviour makes it easier to predict streamflow. The spatial variability in rainfall and temperature is averaged over wide areas, reducing the impact of local errors and data noise. The worst results were

Table 1 | Predictive performance of the three models across basins with only precipitation and temperature as inputs

		NSE			RMSE (m^3/s)			MAE (m^3/s)			MAPE (%)		
Basin	**Model**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
Jardín	MLP	0.56	0.51	0.53	10	10	8	5.0	5.1	5.3	102	106	105
	LSTM	0.45	0.21	0.19	11	12	13	5.0	5.6	6.9	88	57	149
	RF	0.57	0.47	0.43	10	9	11	4.9	5.4	6.3	116	133	139
Paso del Colegio	MLP	0.40	0.36	0.29	115	126	144	91	99	110	22	24	32
	LSTM	0.19	−0.01	0.11	137	102	158	108	83	113	30	19	25
	RF	0.49	0.25	0.22	108	128	132	84	104	106	21	27	28
Puente Balsadero	MLP	0.70	0.70	0.66	69	69	74	48	48	52	22	22	24
	LSTM	0.66	0.61	0.37	88	83	84	56	50	58	23	23	29
	RF	0.71	0.61	0.59	79	91	95	50	58	59	23	27	28
Tablazo	MLP	0.95	0.88	0.89	65	105	90	49	67	63	15	19	19
	LSTM	0.92	0.92	0.82	83	70	143	60	41	100	16	13	18
	RF	0.81	0.69	0.68	131	162	179	80	110	114	23	30	37
Despensa	MLP	0.68	0.64	0.60	116	121	125	78	81	84	28	29	30
	LSTM	0.60	0.53	0.55	129	149	122	91	103	89	40	27	35
	RF	0.72	0.59	0.65	107	139	116	74	91	84	30	37	37

Note: Performance is evaluated for MLP (Multilayer Perceptron), LSTM (Long Short-Term Memory), and RF (Random Forest) models during Train (training), Val (validation), and Test (testing) phases. Results are quantified by NSE (Nash-Sutcliffe Efficiency) for predictive power, RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for absolute error magnitude, and MAPE (Mean Absolute Percentage Error) for relative error.

gained for Paso del Colegio, particularly for LSTM, where the models failed to capture the streamflow variability. This can be explained by the operations of the CHQ reservoir in close proximity upstream of the streamflow gauge, which modulate the rainfall-runoff processes. This basin also has a comparably short temporal record, along with Puente Balsadero, as both basins correspond to rivers transformed by CHQ, and the data start in 2016.

Among the tested approaches, the MLP model demonstrated the most consistent performance across the training, validation, and testing datasets in the Jardín, Paso del Colegio, Puente Balsadero, and Despensa basins. This can be attributed to the fact that overfitting was easier to control in the MLP architecture compared with the other models evaluated. The superiority of MLP is consistent with previous studies. For example, Jesús *et al.* (Jesus *et al.* 2025) reported that MLP models achieved acceptable accuracy in short-term river flow forecasting. Similarly, Mutlu *et al.* (2008) compared two types of ANN models (MLP and RBF) for predicting streamflow at four stations using different lagged inputs. Both models performed satisfactorily; however, the MLP consistently outperformed the RBF network.

However, for Jardín and Paso del Colegio, NSE values remained low, and both relative and absolute errors remained high, even for the best models. This is particularly the case in Paso del Colegio, where NSE was in the order of 0.29–0.40. In this basin, all models struggled to represent the hydrological dynamics with just temperature and precipitation as input variables. This underscores that model performance is highly basin-specific rather than universally transferable. In contrast, the LSTM model was the most consistent and yielded the best performance in Tablazo. This may be related to a longer hydrological memory in this basin, driven by its larger contributing area and the presence of multiple small reservoirs in its tributaries that buffer and attenuate flows, thereby influencing downstream discharge over several days. These findings are consistent with Yokoo *et al.* (2022), who also reported that LSTM can produce acceptable and accurate daily streamflow forecasts using precipitation and air temperature as the only inputs. Hence, under certain basin conditions, LSTM can learn sufficient nonlinear patterns from precipitation and temperature alone to forecast streamflow well. It is also noteworthy that the Tablazo basin had by far the largest number of measurement stations (116 precipitation stations and 38 temperature stations), indicating that the availability of more input data can lead to improved model performance, particularly for the LSTM model.

Figure 2 shows that model performance is strongly conditioned by flow regime and ENSO phase rather than by the machine learning architecture alone. Across all basins, errors increased markedly during high-flow conditions, indicating that extreme inflow events remain the most challenging regime to predict, regardless of the model used. During El Niño conditions, models generally performed well under low and normal flows, particularly in basins such as Jardín and Balsadero, where reduced rainfall variability enhances predictability. In La Niña, performance was poorer when predicting high flows.

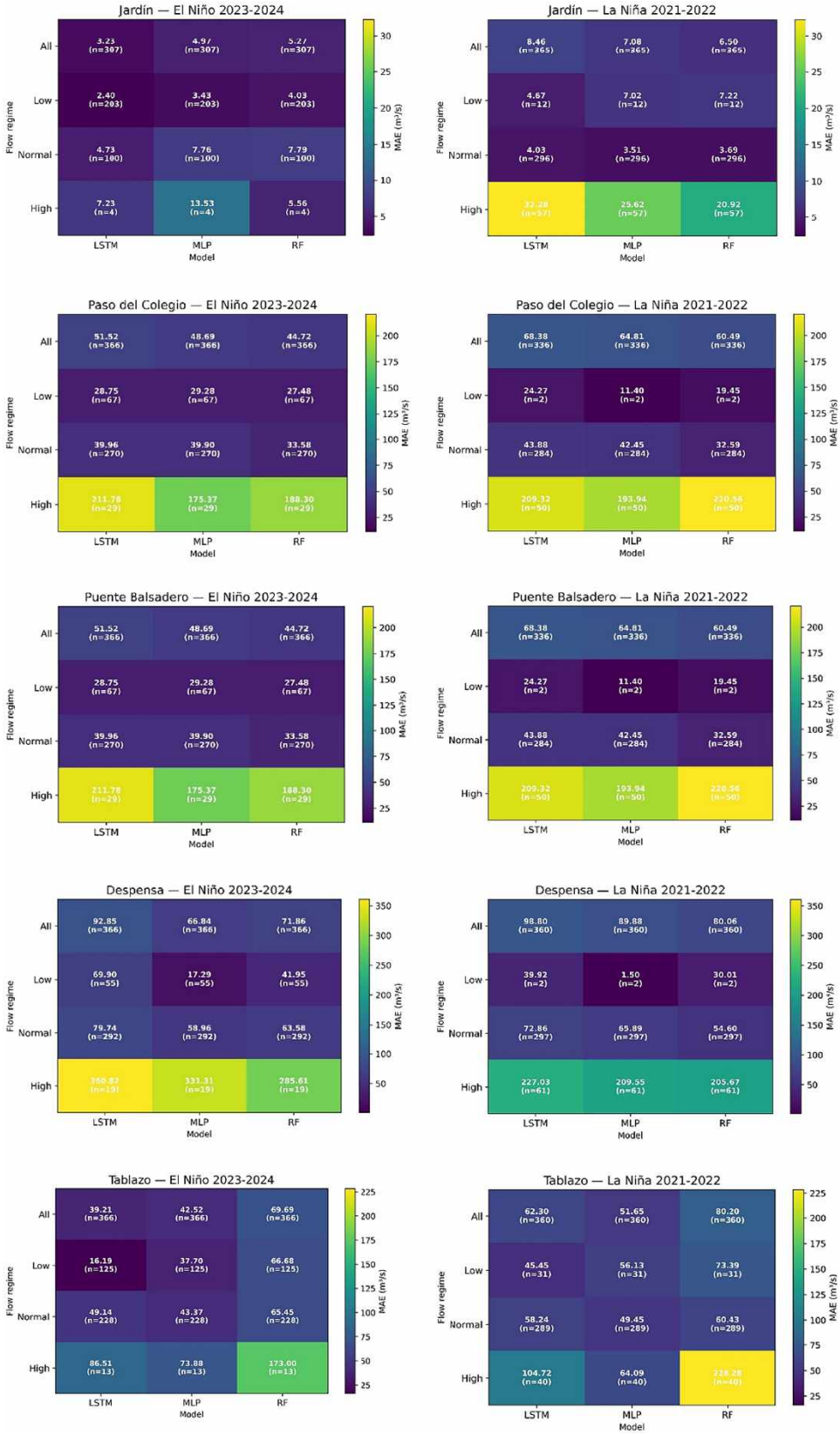


Figure 2 | Mean Absolute Error (MAE) across streamflow regimes (low, normal, and high flows) for the three models (LSTM, MLP, and RF) with precipitation and temperature as inputs.

Especially in rapid-response basins such as Jardin and in highly regulated basins such as Paso del Colegio, the increased frequency and variability of intense rainfall events during La Niña challenged the three models.

Temperature, precipitation and past streamflow as input

The inclusion of previous-day streamflow as an input significantly improved predictive performance and model consistency across all five basins by providing a proxy for antecedent moisture and storage (see Table 2). The Multilayer Perceptron (MLP) is the most robust model, demonstrating superior generalization and achieving the highest predictive skill in basins such as Jardín and Paso del Colegio. While Random Forest (RF) showed marked improvements in consistency and often achieved the lowest relative errors (MAPE), it occasionally failed to capture peak flows as well as MLP did. LSTM performance also improved, particularly in the larger Tablazo basin, yet it continued to struggle with generalization and overfitting, especially in Puente Balsadero.

Despite these gains, capturing the full range of hydrological variability remains a challenge. In basins like Paso del Colegio, while the improvement was the most significant compared to the first configuration, the models still struggled to accurately represent high- and low-flow events. Across most basins, high RMSE values persist, indicating that while adding lagged streamflow data helps the models learn underlying patterns, they still tend to underestimate peak flows.

The results indicate an increase in forecasting capacity for all basins and models when lagged inflows are included as predictors. This improvement, however, is not uniform across basins and reflects differences in hydrological response and system memory. The gain in performance is especially relevant for Paso del Colegio and less critical for Tablazo. For Paso del Colegio, the improvement provides access to a combined signal (hydrological + operational), where the operational component reflects the influence of the CHQ reservoir. This combined signal is not fully captured by precipitation and temperature alone. In Tablazo, due to the good instrumentation of the basin and lower anthropogenic transformation, the improvement is not as pronounced as in Paso del Colegio.

A decrease in RF overfitting is observed in all basins when lagged streamflow is included as an input. This reduction is attributed to the fact that lagged streamflow carries information about the antecedent catchment state, which precipitation and temperature alone cannot represent. When this memory of the basin is provided explicitly, the model does not need to infer this condition from the meteorological signals. This finding is consistent with those of Moges *et al.* (2024), who reported that RF performs better when historical streamflow and time-lagged meteorological inputs are included, compared to using only current meteorological variables.

Table 2 | Predictive performance of the three models across basins with precipitation, temperature and past streamflow as inputs

		NSE			RMSE (m^3/s)			MAE (m^3/s)			MAPE (%)		
Basin	**Model**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
Jardín	MLP	0.73	0.64	0.71	7.9	8.7	6.2	4.0	4.4	3.5	33	38	34
	LSTM	0.68	0.34	0.29	8	10	13	4.1	4.8	6.4	71	55	97
	RF	0.70	0.61	0.63	8	10	10	3.1	3.7	4.0	36	42	44
Paso del Colegio	MLP	0.70	0.64	0.61	80	94	106	63	71	76	15	17	22
	LSTM	0.51	0.23	0.41	107	89	129	84	72	94	22	17	22
	RF	0.70	0.61	0.60	81	97	107	60	70	78	14	17	23
Puente Balsadero	MLP	0.77	0.75	0.75	60	63	64	46	43	45	20	19	20
	LSTM	0.91	0.66	0.50	43	73	75	32	44	50	16	22	25
	RF	0.79	0.77	0.74	70	60	66	40	40	44	16	18	19
Tablazo	MLP	0.96	0.90	0.90	60	97	86	44	60	59	13	16	17
	LSTM	0.91	0.93	0.87	87	62	124	65	40	86	17	15	16
	RF	0.90	0.83	0.88	93	126	96	51	64	54	12	15	14
Despensa	MLP	0.83	0.81	0.83	85	89	82	51	52	53	16	17	17
	LSTM	0.68	0.66	0.70	115	125	100	74	81	69	26	19	22
	RF	0.85	0.80	0.81	80	91	85	46	52	54	14	16	17

Note: Performance is evaluated for MLP (Multilayer Perceptron), LSTM (Long Short-Term Memory), and RF (Random Forest) models during Train (training), Val (validation), and Test (testing) phases. Results are quantified by NSE (Nash-Sutcliffe Efficiency) for predictive power, RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for absolute error magnitude, and MAPE (Mean Absolute Percentage Error) for relative error.

As in the first configuration, MLP remains the most consistent model across Jardín, Paso del Colegio, and Despensa, and now also in Tablazo. In Puente Balsadero, however, RF emerges as the most consistent model.

Regarding ENSO events, Figure 3 shows that including lagged inflow enhanced daily streamflow predictability by reinforcing hydrological memory, leading to robust performance under low and normal flow conditions and improved high flow performance. However, these events remained difficult to forecast across basins. Variations among basins suggest that prediction constraints during high-flow events are predominantly influenced by basin-specific hydrological dynamics and anthropogenic infrastructure, rather than by the selection of a specific model. In general, it can be shown that including past streamflow values improved model accuracy, particularly in basins where models previously performed poorly.

Nonetheless, this approach introduces greater uncertainty in forecasts of future behaviour, as future past-inflows must be substituted with previous model predictions, leading to the propagation of prediction errors into the future. Whether this increase in uncertainty is justified by the improvement in model accuracy ultimately depends on the specific use case.

Monthly data

Temperature and precipitation as input

The daily data were aggregated to a monthly time scale to examine whether this temporal resolution could lead to an overall improvement in model skill, particularly for peak-flow forecasting. This third model configuration uses monthly average temperature and precipitation to forecast streamflow. Table 3 summarizes the corresponding performance metrics.

The transition from daily to monthly temporal resolution reveals a complex trade-off between noise reduction and data scarcity. By aggregating temperature and precipitation into monthly averages, the models aim to capture larger seasonal trends, though this comes at the cost of a significantly smaller training dataset.

In Jardín and Paso del Colegio, the monthly configuration yielded a significant improvement in forecast accuracy and a reduction in relative error compared to the first configuration. The aggregation process tends to smooth out daily fluctuations, allowing models to better identify long-term rainfall-runoff relationships. However, this benefit did not apply to all basins; in the Tablazo, performance deteriorated compared to the first daily configuration. This decline is likely due to the limited number of monthly observations, which may not provide sufficient information for the models to learn complex dynamics properly.

The MLP was the most reliable architecture at the monthly scale, maintaining the highest overall consistency across the Jardín, Puente Balsadero, Tablazo, and Despensa basins. In Despensa, the MLP demonstrated a superior ability to capture streamflow variability and peak flows, outperforming both previous daily configurations. Moreover, MLP showed mild overfitting in Puente Balsadero and Tablazo, where its predictive accuracy declined during testing.

In the monthly analysis, there is an increasing trend of overfitting due to a smaller sample size. The LSTM network showed an improvement in Paso del Colegio, becoming the most consistent model configuration. However, the other basins experience a decrease in performance. In Puente Balsadero, for instance, the LSTM's testing NSE dropped to negative values, indicating a total loss of predictive skill. This suggests that with fewer data points, complex models like LSTM are more prone to memorizing the training data rather than learning generalizable patterns.

The ability to forecast peak flows at a monthly resolution varied significantly across models and basins. In Despensa and Puente Balsadero, the MLP get a satisfactory peak forecasting capabilities and the best MAPE performance. In Jardín, both the LSTM and RF models struggled significantly to predict high-flow events. While RF generally showed improved consistency compared to the first configuration in most basins, it rarely outperformed the second configuration (daily with lagged flow), suggesting that temporal aggregation does not always compensate for the loss of antecedent moisture data.

Temperature, precipitation, and past inflows as input

Past streamflow was also included for the monthly forecasting. Table 4 summarizes the evaluation metrics.

Overall, including previous streamflow as an additional predictor did not lead to a consistent improvement across all models at the monthly scale. The gains are much less evident than those observed when past streamflow was incorporated into the daily datasets. With monthly data, models can more accurately predict peak flows, reducing the relative impact of previous streamflow on performance. Consequently, this fourth configuration did not yield consistent improvements and, in several instances, increased the risk of overfitting, indicating that it captures patterns specific to the training period rather than learning clear and transferable hydrological relationships.

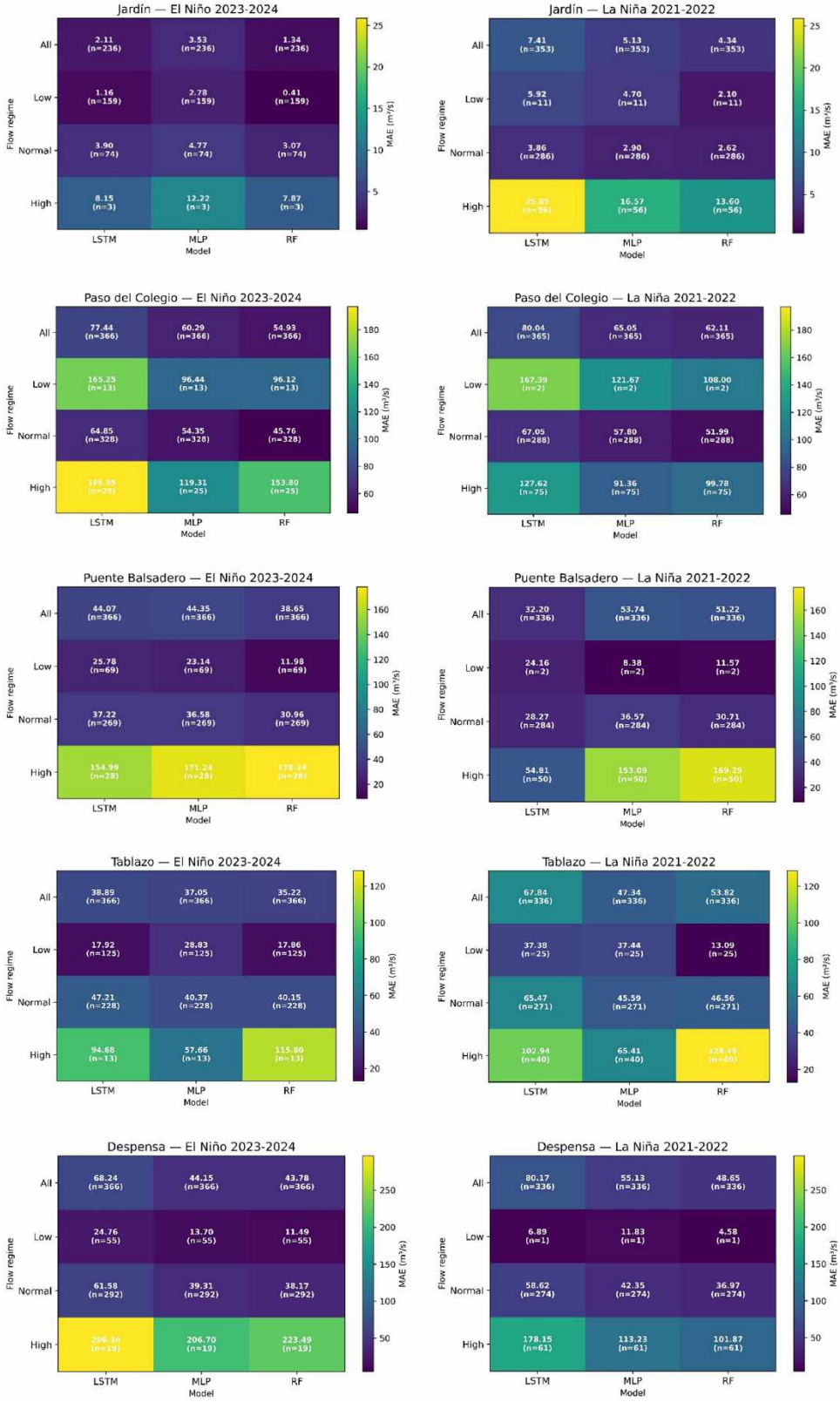


Figure 3 | Mean Absolute Error (MAE) across streamflow regimes (low, normal, and high flows) for the three models (LSTM, MLP, and RF) with precipitation, temperature and past streamflow as inputs.

Table 3 | Predictive performance of the three models across basins with precipitation and temperature at a monthly resolution as inputs

		NSE			RMSE (m^3/s)			MAE (m^3/s)			MAPE (%)		
Basin	**Model**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
Jardín	MLP	0.77	0.74	0.83	4	5	4	3.0	3.4	2.3	58	45	40
	LSTM	0.45	0.22	−0.29	7	5	10	4.4	3.8	7.3	48	54	127
	RF	0.60	0.59	0.78	6	6	4	3.5	4.3	3.1	69	64	81
Paso del Colegio	MLP	0.88	0.46	0.59	37	57	61	26	45	47	6	10	11
	LSTM	0.82	0.60	0.67	44	43	59	35	34	55	8	7	11
	RF	0.44	0.37	0.42	80	56	72	62	45	58	14	9	13
Puente Balsadero	MLP	0.85	0.77	0.57	31	44	61	26	37	52	14	16	15
	LSTM	0.87	0.34	−2.98	32	59	150	25	48	114	11	28	67
	RF	0.63	0.46	0.25	52	57	96	41	50	75	19	25	28
Tablazo	MLP	0.85	0.92	0.63	90	70	83	68	56	68	19	14	18
	LSTM	0.66	0.40	0.43	128	156	187	91	131	151	21	37	30
	RF	0.67	0.48	0.44	131	177	121	88	147	97	29	41	25
Despensa	MLP	0.92	0.90	0.92	45	55	50	35	43	37	15	16	15
	LSTM	0.86	0.76	0.50	58	88	106	47	70	83	18	19	39
	RF	0.83	0.80	0.89	66	76	57	51	64	49	20	29	25

Note: Performance is evaluated for MLP (Multilayer Perceptron), LSTM (Long Short-Term Memory), and RF (Random Forest) models during Train (training), Val (validation), and Test (testing) phases. Results are quantified by NSE (Nash-Sutcliffe Efficiency) for predictive power, RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for absolute error magnitude, and MAPE (Mean Absolute Percentage Error) for relative error.

Table 4 | Predictive performance of the three models across basins with precipitation, temperature and past streamflow at monthly resolution as inputs

		NSE			RMSE (m^3/s)			MAE (m^3/s)			MAPE (%)		
Basin	**Model**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
Jardín	MLP	0.83	0.72	0.39	4	5	9	2.7	3.0	5.5	41	38	59
	LSTM	0.56	0.13	0.14	6	7	8	4.6	5.3	6.2	90	70	205
	RF	0.69	0.62	0.57	5	6	8	3.2	3.6	4.2	57	67	64
Paso del Colegio	MLP	0.68	0.71	0.44	56	60	50	44	46	45	10	10	10
	LSTM	0.81	0.58	0.58	44	43	66	33	37	56	7	7	11
	RF	0.47	0.37	0.48	78	56	68	61	44	56	13	9	13
Puente Balsadero	MLP	0.71	0.68	0.73	18	44	53	49	32	36	15	19	11
	LSTM	0.68	0.32	−2.53	52	60	134	38	48	99	15	29	68
	RF	0.64	0.44	0.31	52	58	92	41	50	72	19	26	26
Tablazo	MLP	0.97	0.91	0.76	41	74	67	34	59	61	10	14	19
	LSTM	0.65	0.76	0.69	132	91	169	94	76	132	19	40	22
	RF	0.67	0.47	0.47	131	179	118	86	148	93	29	41	24
Despensa	MLP	0.93	0.85	0.88	44	60	58	34	48	44	13	17	16
	LSTM	0.81	0.74	0.72	68	95	78	54	69	68	20	16	27
	RF	0.82	0.78	0.90	68	72	51	55	57	36	23	20	11

Note: Performance is evaluated for MLP (Multilayer Perceptron), LSTM (Long Short-Term Memory), and RF (Random Forest) models during Train (training), Val (validation), and Test (testing) phases. Results are quantified by NSE (Nash-Sutcliffe Efficiency) for predictive power, RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) for absolute error magnitude, and MAPE (Mean Absolute Percentage Error) for relative error.

In basins like Jardín and Paso del Colegio, adding previous streamflow did not improve the performance of the model compared to previous configurations. In Jardín, the MLP struggled with generalization, while the LSTM continued to exhibit severe overfitting, confirming it was not a viable model for that basin. Similarly, in Paso del Colegio, while the RF and MLP showed some improvements over the first configuration, they could not match the superior consistency and performance achieved in the second configuration (daily with lagged flow).

Performance in Puente Balsadero and Tablazo was characterized by moderate to high instability. In Puente Balsadero, while the MLP maintained good consistency, the LSTM and RF models showed unsatisfactory behaviour, with negative test metrics and significant overfitting. In Tablazo, although the MLP improved during training, its decline during the testing phase made it less consistent than the daily configurations. Meanwhile the RF model's performance was notably poor, having the same difficulties as in the third configuration.

The Despensa basin was the only basin where the fourth configuration had really good results. In this basin, all three models (MLP, LSTM, and RF) demonstrated high performance and good generalization skills. Both the MLP and RF outperformed their daily configurations, indicating that, for this basin (one of those with the longest available records of precipitation, streamflow, and temperature), the combination of extended data availability, monthly aggregation, and antecedent flow information helps improve the representation of overall streamflow variability.

Across the tested monthly scenarios, the Multilayer Perceptron (MLP) remained the most adaptable architecture, despite its occasional problems with generalization in smaller basins. The LSTM proved particularly sensitive at this scale, often failing to generalize unless the specific basin characteristics favored its memory-based structure. Overall, the results suggest that the added complexity of lagged inputs at the monthly level often leads to worse results, as the models struggle to maintain consistency given the limited number of monthly data points.

DISCUSSION

The comparison between daily and monthly resolutions shows a trade-off between capturing detail and maintaining consistency. Daily resolution is important for representing the basin's rapid responses, but it is also noisier, which can lead to higher percentage errors. Monthly data acts as a filter, smoothing out these daily changes and making it easier to find long-term seasonal trends, but it also leaves the models with considerably less data to learn from, making them prone to memorize specific data rather than learning the actual hydrological relationships. Adding previous streamflow informs the model about the basin's current wetness (soil moisture and storage), which is important for predicting high-water peaks in daily models. At the monthly level, this is less helpful because the average flow already reflects much of that seasonal memory. However, the models still fail to capture high peaks, especially in La Niña years, where these events are more frequent.

In addition to these physical limitations, the effectiveness of the monthly approach is constrained by the reduced size of the training datasets. When lagged inflow is included as an additional predictor, monthly aggregation does not consistently improve performance across all basins. In this case, the number of available data points is reduced because missing streamflow records were not filled in order to avoid introducing additional uncertainty into the modeling framework. Consequently, fewer samples are available for model training, validation, and testing at this configuration, which can limit model generalization. This effect is particularly evident in basins associated with relatively recent hydropower developments, such as those influenced by the CHQ and CHS reservoirs, where shorter streamflow records constrain predictive performance.

Consequently, model reliability is not only a function of record length but is also deeply influenced by specific basin characteristics, such as instrumentation density and drainage area. Large, well-instrumented basins like Tablazo provide a clear signal that enables complex models like LSTM to perform well, whereas small, rapid-response basins require a simpler model, such as an MLP, to perform well. Furthermore, human intervention, such as the El Quimbo operations near Paso del Colegio, adds further complexity to the models, as these operations disrupt the usual relationship between rainfall and river flow.

The observed performance differences can be understood by considering the variation of hydrological processes and other characteristics across the basins, especially the nature of tropical precipitation and runoff generation. Tropical rainfall is predominantly convective, characterized by high intensities, intermittency, and pronounced spatial variability, with large precipitation contrasts over short horizontal distances (in the order of a few kilometers). Tropical rainfall also exhibits strong temporal variability associated with short-lived, episodic rather than continuous events. These characteristics, combined with rapid runoff generation, increase variability in daily streamflow records and reduce predictability. Under such conditions, models relying exclusively on meteorological inputs tend to exhibit lower skill at the daily scale, particularly in small, fast-responding basins such as Jardín, where rapid runoff generation limits the persistence of hydrological signals. In contrast, larger basins like Tablazo integrate heterogeneous climatic and physiographic conditions across space, thereby weakening the direct linkage between localized precipitation and downstream discharge. The larger scale often provides a

hydrological buffering effect that, when coupled with dense instrumentation, makes the long-term patterns in large basins more predictable than the erratic peaks of smaller catchments. Similarly, in highly intervened systems such as Paso del Colegio, where streamflow integrates heterogeneous hydrological responses and human interventions, mostly related to the operations of the El Quimbo dam upstream, the rainfall–runoff relationship cannot be fully captured by meteorological inputs alone. In these cases, including previous streamflow information increases predictability by implicitly accounting for basin memory, antecedent moisture conditions, and unobserved storage dynamics, thereby reducing uncertainty in initial conditions and enhancing the representation of both fast runoff and slow baseflow processes.

From an operational perspective, this limitation further reinforces the practical relevance of daily-scale forecasting. Hydropower operators require forecasts at short temporal horizons because operational decisions are made at the hourly scale. Reservoir management requires deciding hourly how much water to release, store, or route through spillways, and operational risk is primarily associated with extreme inflow events rather than with monthly mean conditions. Anticipating daily inflows to the reservoir enables operators to plan plant operations for the following day; for example, if a high-flow event is expected while the reservoir is near full capacity, preventive measures can be implemented, including controlled gate openings and early warnings to downstream communities to mitigate the risk of sudden water level rises and potential flooding.

To place these findings in a broader hydrological context, the results of this study are compared with those reported in studies using physically based or conceptual hydrological models in Colombian tropical basins. Results from recent studies applying physically based hydrological models in Colombian river basins show a wide range of performance when evaluated using NSE. In the tropical mountainous basin of the Sierra Nevada de Santa Marta, the SWAT model yielded high NSE values during both calibration (0.72) and validation (0.79), enabling reliable assessment of streamflow response under persistent drought scenarios (Hoyos *et al.* 2019). The SAC-SMA model, in turn, when compared across 12 Andean basins characterized by strong climatic and geomorphological variability, showed substantial heterogeneity in model performance across basins (Alvarez & Barco 2025). For example, in the La Miel basin, NSE values during calibration and validation exceeded 0.7, whereas in other basins, such as the Sumapaz River, NSE values remained below 0.5, reflecting differences in hydrological response, data availability, and basin complexity. This basin-dependent variability is consistent with the patterns observed in the present study, where model performance varies markedly across catchments. Additionally, Alvarez & Barco (2025) reported that the SAC-SMA model encounters difficulties in accurately capturing high flows, a limitation that is likewise observed in the present study. In this context, the performance levels obtained in the present study fall within the range commonly reported for physically based models in Colombian tropical basins, while relying on a substantially reduced set of input variables.

While the performance of the machine learning models was previously compared with results from physically based models, an additional question arises: whether removing the highest precipitation percentile could have affected model performance. To investigate this possibility, the authors re-ran the models for the three basins with the lowest performance metrics (Jardín, Despensa, and Paso del Colegio) using the initial configuration (daily resolution with precipitation and temperature inputs), but without filtering the upper precipitation outliers. The results showed no improvement in model performance. Instead, including precipitation values above the 99.5th percentile introduced substantial noise, leading to overfitting, particularly in the MLP and LSTM architectures, without improving the representation of peak flows. These findings suggest that the removed values were more likely due to localized measurement errors rather than to meaningful climatic extremes, supporting the use of the 99.5th percentile threshold as a quality-control step to maintain model robustness and generalization in these tropical basins.

CONCLUSIONS

Overall, comparing three reservoir inflow forecasting models with two sets of predictors at two temporal resolutions on five basins shows that no single configuration or temporal resolution performs best for all basins. Model performance varies systematically across basins, reflecting differences in hydrological response and basin-specific conditions rather than differences in model structure alone. In some cases, precipitation and temperature alone yield sufficiently accurate forecasts, as in the Tablazo and Despensa basins. In other basins, incorporating past streamflow to represent antecedent conditions is necessary to improve predictive skill, as in Paso del Colegio. Incorporating past streamflow as an additional predictor yielded clear benefits for all basins at daily resolution, with particularly strong improvements observed in three of the basins. However,

at the monthly scale, the effect is less pronounced: in some cases, the improvement is modest, and in others, model performance even deteriorates. This is likely because, at a monthly resolution, memory effects are largely captured by the timestep itself. Furthermore, the inclusion of lagged streamflow significantly reduced the available dataset; since missing records were not filled, each gap in the daily series resulted in the loss of multiple subsequent training samples due to the lagging window. This reduction in data points particularly constrained the performance of the models.

In basins with comparably short data records, aggregating the daily information to monthly values tended to increase overfitting, despite improvements in peak-flow prediction. Hence, monthly aggregation does not necessarily guarantee better overall model robustness under limited data scenarios. Moreover, in more heavily instrumented basins, LSTM models tend to outperform other architectures when sufficient data are available. In contrast, in smaller basins with sparse monitoring networks or high levels of human intervention, simpler models such as MLP often achieve better performance.

This study contributes to hydropower forecasting by clarifying when and why certain predictors and temporal resolutions are more effective. In large, well-instrumented basins with limited human intervention, temperature and precipitation alone are often sufficient to achieve good predictive performance. In contrast, in smaller and more heavily regulated basins, it becomes necessary to include previous inflows and, potentially, additional variables not considered in this study to further improve forecast skill. Moreover, daily-resolution data tend to provide richer information and are more useful for the daily operation of hydropower plants. When data availability is limited, however, monthly resolution is not advisable, as data-driven models require large datasets to perform reliably. These insights have practical implications for reservoir operation, as they can guide the selection of forecasting strategies that better support decision-making under data limitations and climate variability. Future research would benefit from integrating machine learning with physically based representations or ensemble techniques, to improve model stability and enhance inflow forecasting skill in complex tropical basins.

FUNDING

This work was funded by the Volkswagen Foundation through grant no. 96955 (Water Security for Whom?).

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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First received 4 December 2025; accepted in revised form 11 March 2026. Available online 27 March 2026